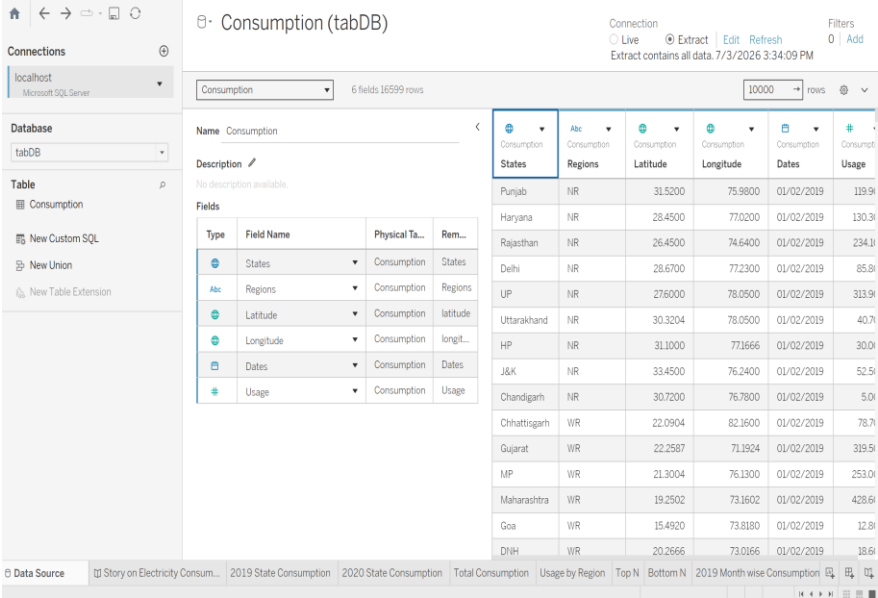


Project Performance Test

Date	15 July 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	

Model Performance Testing:

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	 The electricity consumption dataset was successfully imported into Tableau Desktop and rendered for analysis. The dataset contains fields such as States, Regions, Latitude, Longitude, Dates, and Usage, which serve as the foundation for creating interactive dashboards and visualizations. The imported data was verified to ensure that all records were loaded correctly before dashboard development.

2. Data Preprocessing

Consumption (tabDB)

Connection: Live | Extract | Edit | Refresh | Filters: 0 | Add

Extract contains all data: 7/3/2026 3:34:09 PM

Consumption 6 fields 36599 rows 10000 rows

Name: Consumption

Description: No description available.

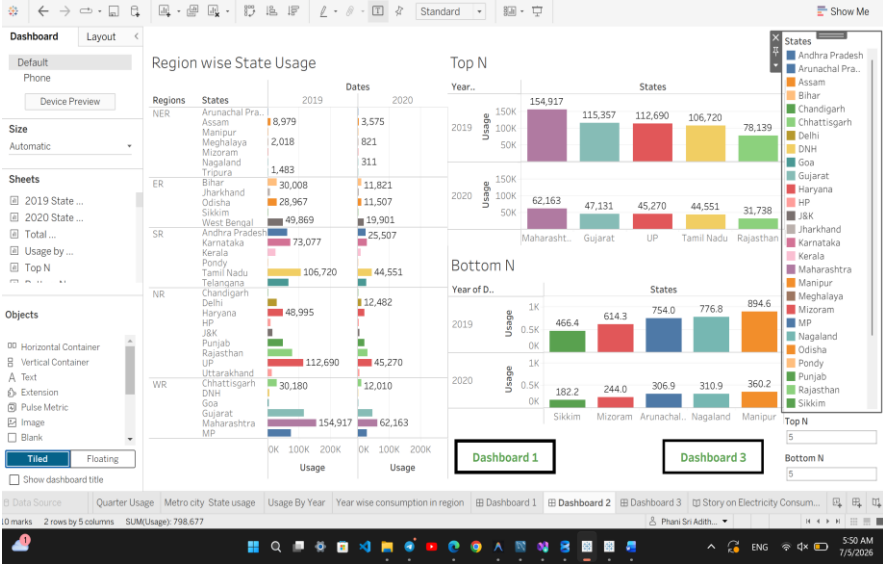
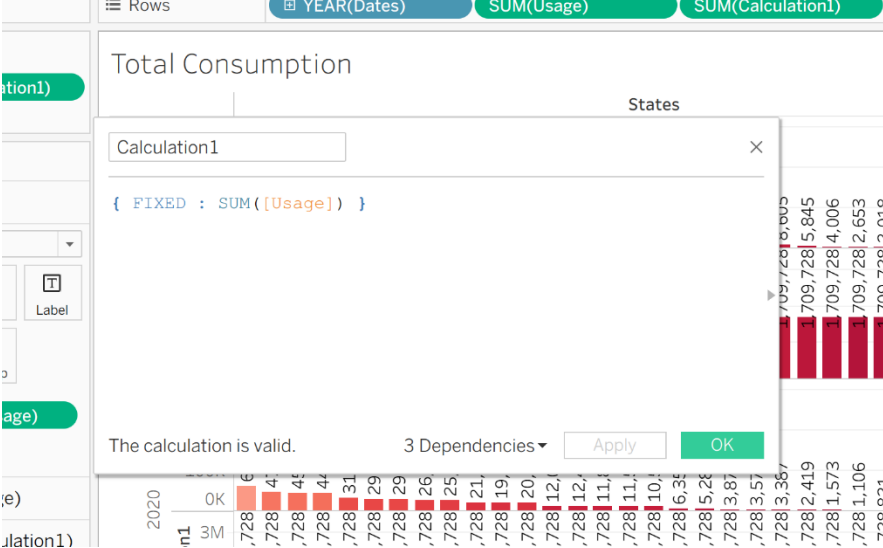
Fields:

Type	Field Name	Physical Ta...	Rem...
Dimension	States	Consumption	States
Dimension	Regions	Consumption	Regions
Dimension	Latitude	Consumption	Latitude
Dimension	Longitude	Consumption	Longitude
Dimension	Dates	Consumption	Dates
Measure	Usage	Consumption	Usage

Consumption	Abc	Consumption	Consumption	Consumption	Consumption
States	Regions	Latitude	Longitude	Dates	Usage
Punjab	NR	31.5200	75.9800	01/02/2019	119.9
Haryana	NR	28.4500	77.0200	01/02/2019	130.3
Rajasthan	NR	26.4500	74.6400	01/02/2019	234.1
Delhi	NR	28.6700	77.2300	01/02/2019	85.8
UP	NR	27.6000	78.0500	01/02/2019	313.9
Uttarakhand	NR	30.3204	78.0500	01/02/2019	40.7
HP	NR	31.1000	77.3666	01/02/2019	30.0
J&K	NR	33.4500	76.2400	01/02/2019	52.5
Chhattisgarh	NR	30.7200	76.7800	01/02/2019	5.0
Chhattisgarh	WR	22.0904	82.3600	01/02/2019	78.7
Gujarat	WR	22.2587	71.1924	01/02/2019	319.5
MP	WR	21.3004	76.1300	01/02/2019	253.0
Maharashtra	WR	19.2502	73.3602	01/02/2019	428.6
Goa	WR	15.4920	73.8180	01/02/2019	12.8
DH	WR	20.2666	73.0166	01/02/2019	18.6

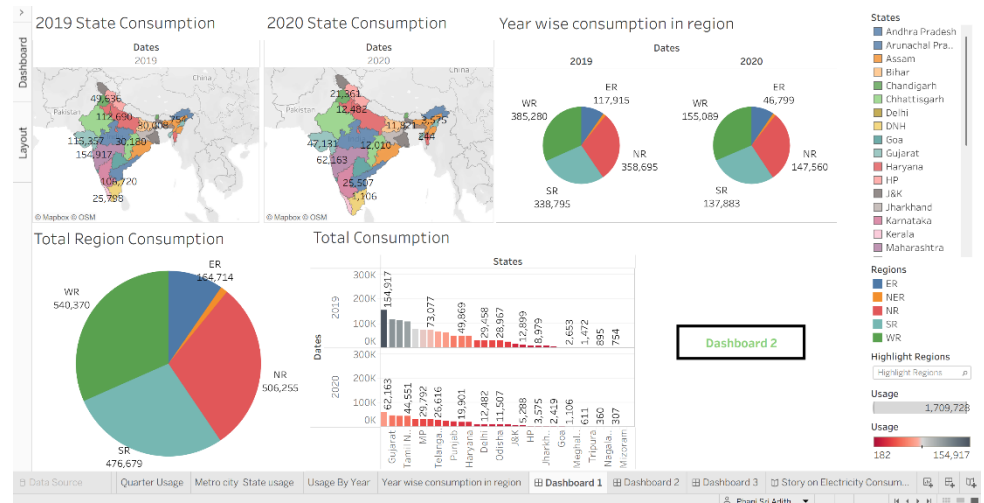
Data Source | Story on Electricity Consum... | 2019 State Consumption | 2020 State Consumption | Total Consumption | Usage by Region | Top N | Bottom N | 2019 Month wise Consumption

The imported dataset was preprocessed by validating field data types, checking for missing or inconsistent values, and organizing the data into appropriate dimensions and measures. A Tableau data extract was created to improve dashboard performance and enable efficient analysis. The prepared dataset was then used to build interactive visualizations, dashboards, and stories.

3. Utilization of Filters	 Interactive filters and parameters were implemented in Tableau to enhance data exploration and dashboard usability. Users can filter the visualizations by State and adjust Top N and Bottom N values to analyze the highest and lowest electricity-consuming states. These filters enable dynamic comparisons, improve dashboard responsiveness, and provide customized insights into regional electricity consumption patterns.
4. Calculation fields Used	

No of Visualizations / Graphs – 13

Figure 5.1: Dashboard 1 – State-wise and Regional Electricity Consumption Analysis

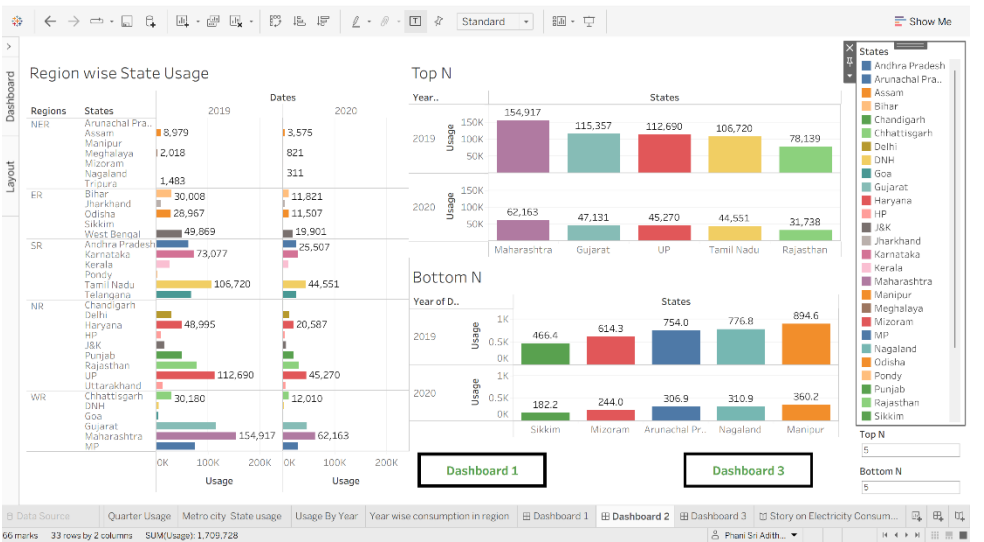


Dashboard 1 provides a comprehensive overview of electricity consumption across Indian states for **2019** and **2020**. It includes two geographical maps displaying state-wise electricity consumption, pie charts showing regional electricity distribution, and a bar chart comparing the total electricity consumption of each state. Interactive filters for **State**, **Region**, and **Usage** enable users to perform dynamic comparisons and identify regional consumption patterns.

Visualizations Included : 6

- State-wise Electricity Consumption Map (2019)
- State-wise Electricity Consumption Map (2020)
- Region-wise Consumption Pie Chart (2019)
- Region-wise Consumption Pie Chart (2020)
- Total Region Consumption Pie Chart
- Total State Consumption Bar Chart

Figure 5.2: Dashboard 2 – Regional Comparison with Top and Bottom State Analysis

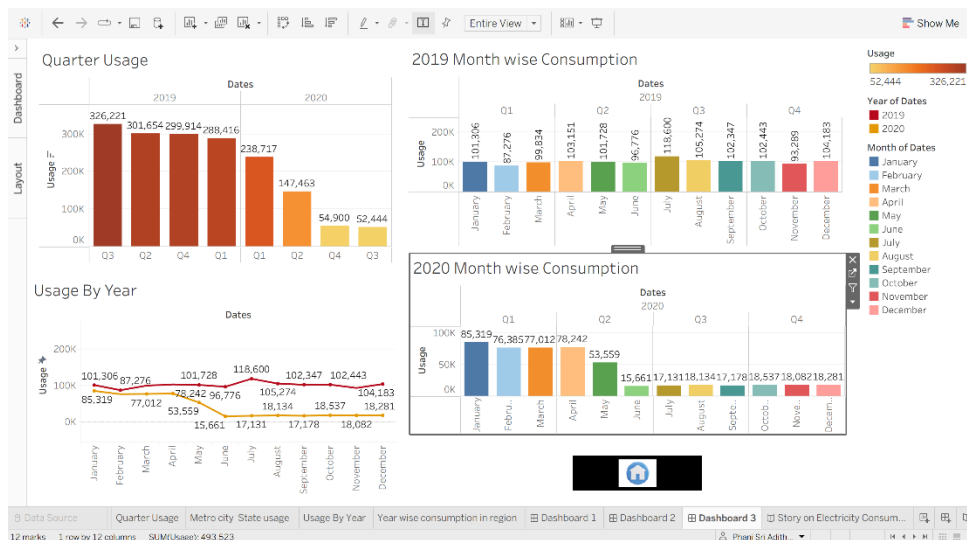


Dashboard 2 focuses on comparative analysis of electricity consumption across Indian regions. It presents a region-wise state consumption chart alongside **Top N** and **Bottom N** state comparisons for both **2019** and **2020**. Interactive parameters allow users to dynamically modify the number of highest and lowest electricity-consuming states displayed, providing deeper analytical insights.

Visualizations Included : 3

- Region-wise State Usage
- Top N States (2019 & 2020)
- Bottom N States (2019 & 2020)

Figure 5.3: Dashboard 3 – Quarterly, Monthly, and Yearly Consumption Trend Analysis

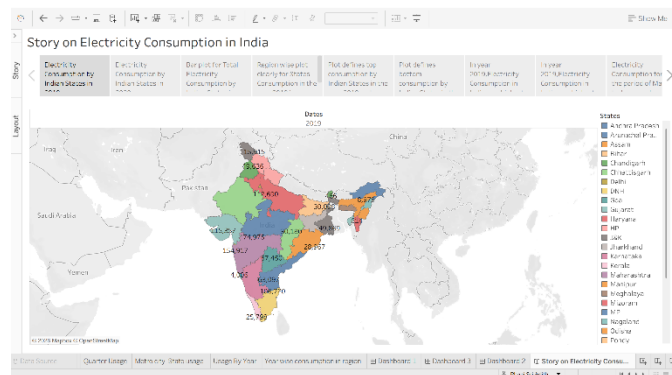


Dashboard 3 analyzes electricity consumption trends over time using quarterly, monthly, and yearly comparisons. It enables users to observe seasonal variations, annual consumption patterns, and the impact of the COVID-19 period on electricity demand. Interactive filters for **Year**, **Month**, and **Usage** support detailed time-based exploration.

Visualizations Included : 4

- Quarter-wise Electricity Consumption
- Monthly Consumption (2019)
- Monthly Consumption (2020)
- Year-wise Electricity Consumption Trend

No of Visualizations / Graphs – 12



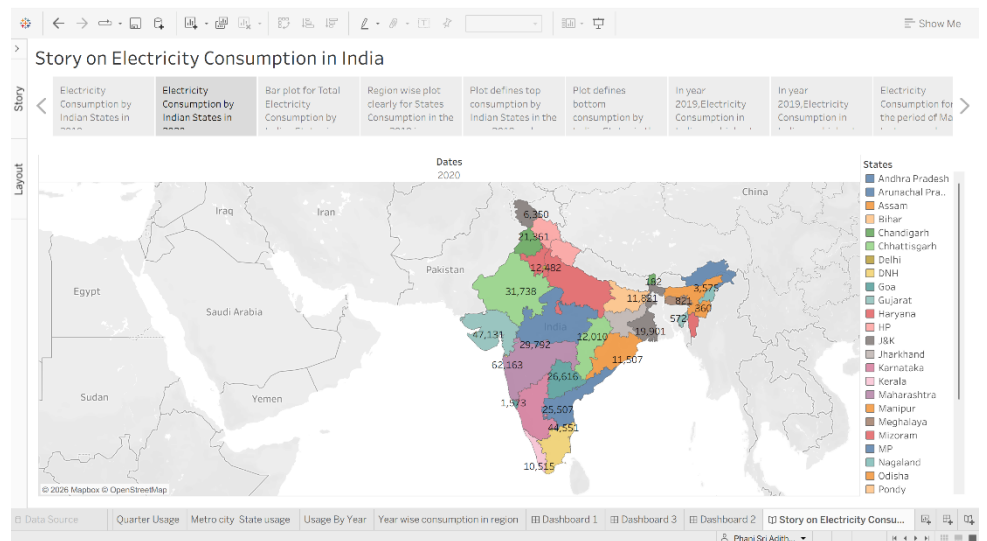
Story Point 1

Title

Electricity Consumption by Indian States in 2019

Description

This story displays the electricity consumption of all Indian states during 2019 using a geographical map. It helps identify states with high and low electricity usage before the COVID-19 pandemic.



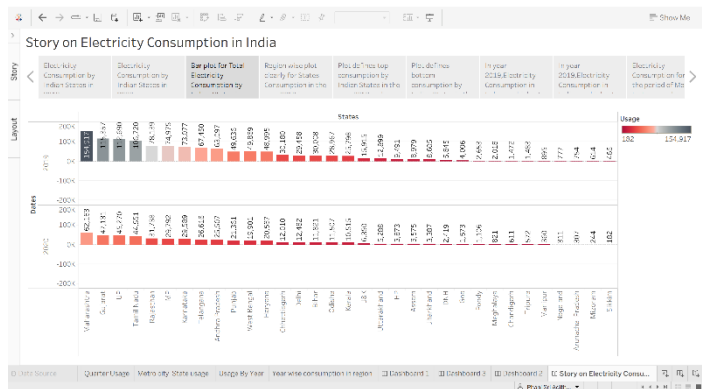
Story Point 2

Title

Electricity Consumption by Indian States in 2020

Description

This map visualizes electricity consumption across Indian states during 2020. It highlights the significant reduction in demand due to the nationwide lockdown and changing consumption patterns.



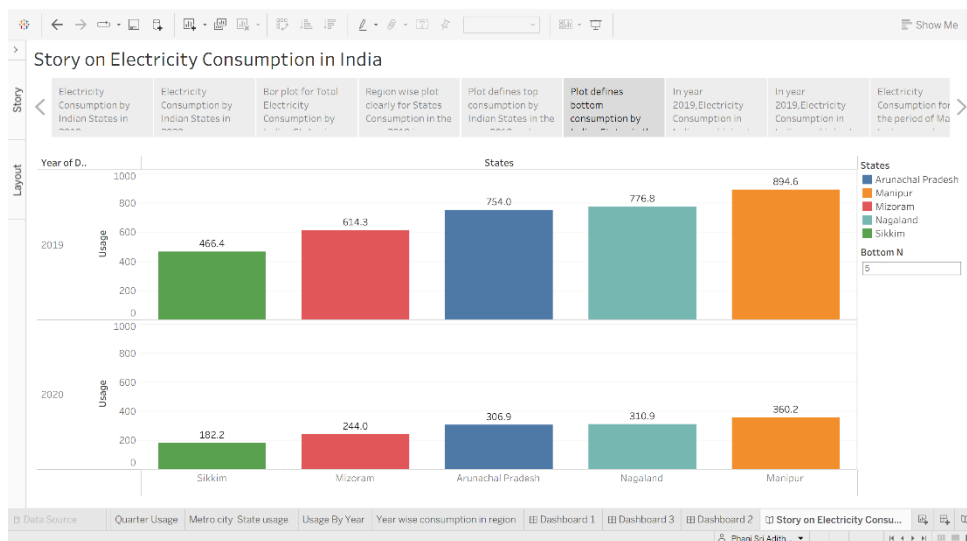
Story Point 3

Title

Total Electricity Consumption by States

Description

This visualization compares the total electricity consumption of all states for 2019 and 2020. It enables users to compare annual electricity usage across states and observe the decline in consumption during the pandemic.



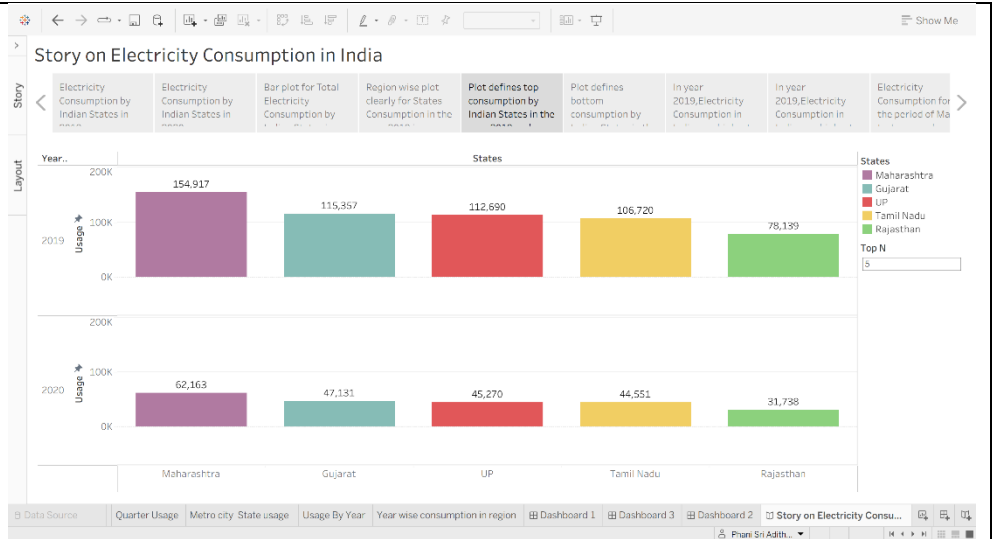
Story Point 4

Title

Bottom 5 Electricity Consuming States

Description

This chart displays the five states with the lowest electricity consumption during both years. It helps identify regions with comparatively lower electricity demand.



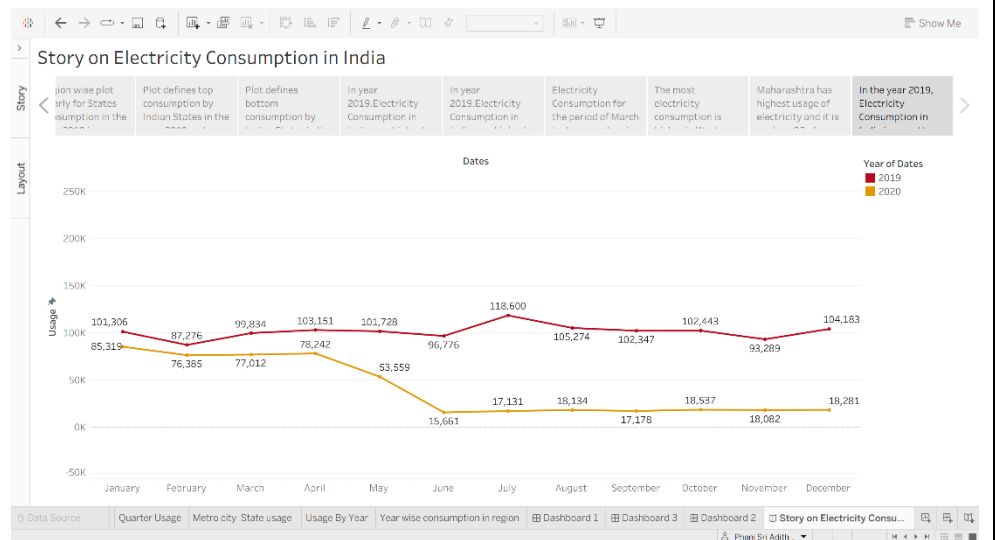
Story Point 5

Title

Top 5 Electricity Consuming States

Description

This visualization highlights the five states with the highest electricity consumption in both 2019 and 2020. Maharashtra remained the highest electricity-consuming state throughout the analysis period.



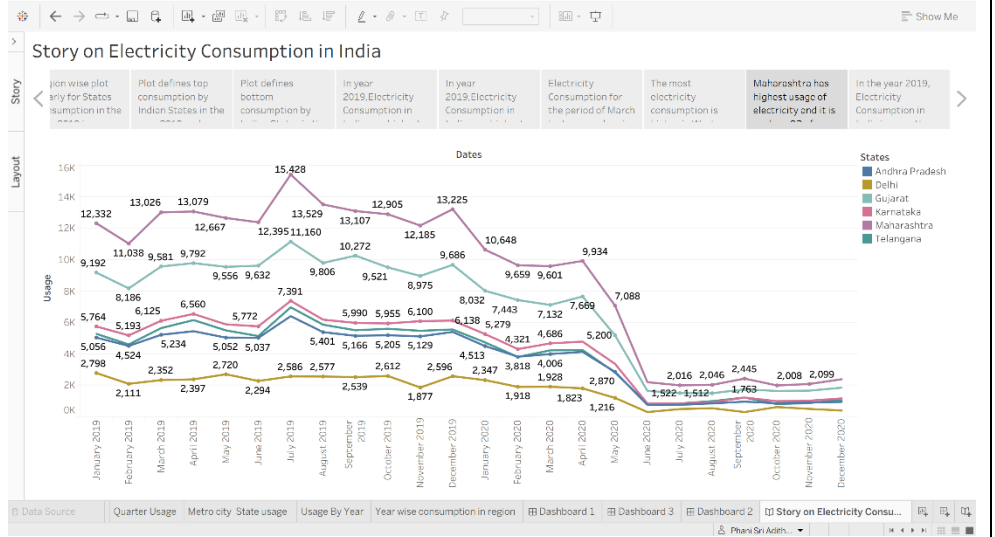
Story Point 6

Title

Overall Electricity Consumption Trend (2019 vs 2020)

Description

This line chart compares monthly electricity consumption between 2019 and 2020. It highlights the overall trend, the sharp decline during lockdown, and the gradual recovery in the later months of 2020.



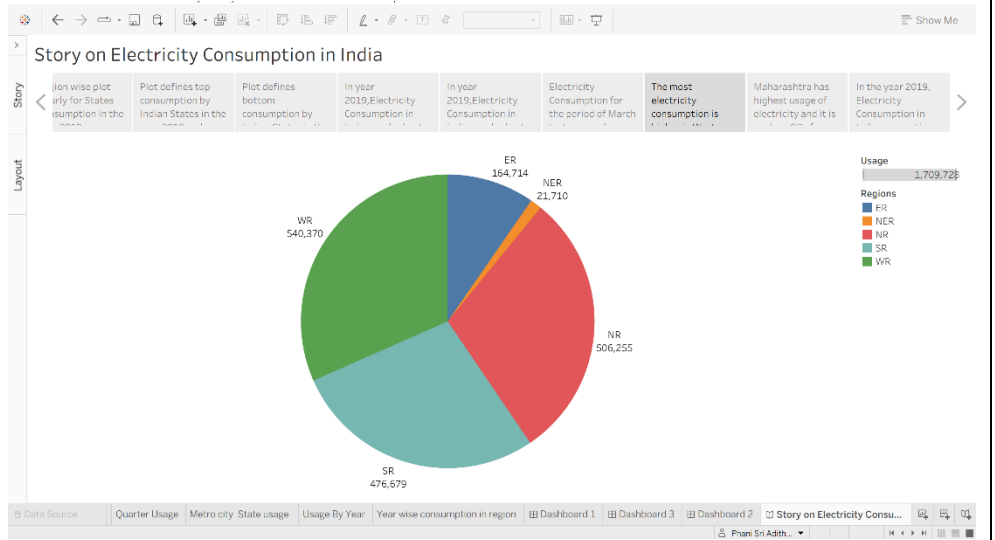
Story Point 7

Title

Maharashtra – Highest Electricity Consuming State

Description

This story focuses on Maharashtra and compares its electricity consumption with other major states over time. The visualization confirms that Maharashtra consistently recorded the highest electricity consumption, reflecting its strong industrial and commercial activity.



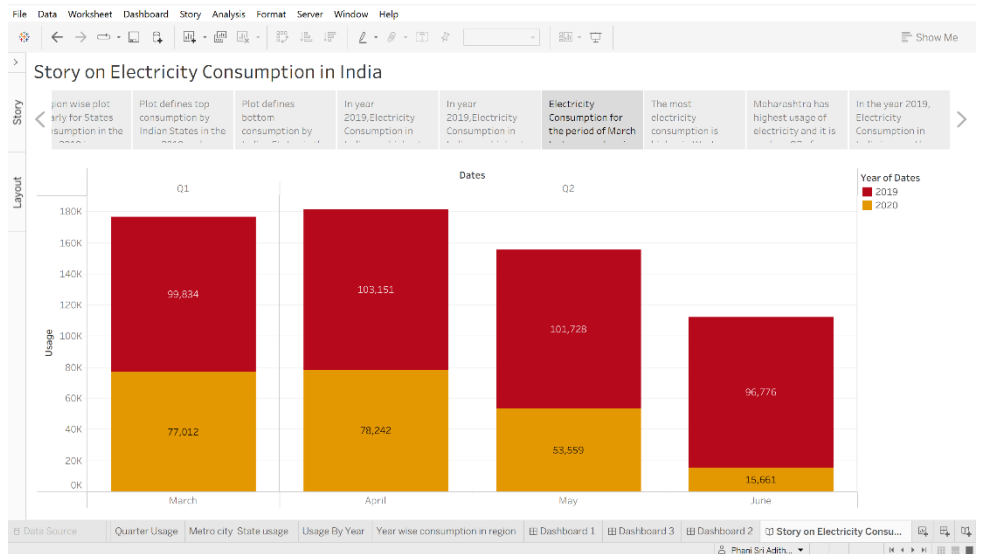
Story Point 8

Title

Region-wise Electricity Consumption

Description

This story presents electricity consumption grouped by geographical regions of India. It compares the contribution of each region in both years and helps identify regional demand patterns.



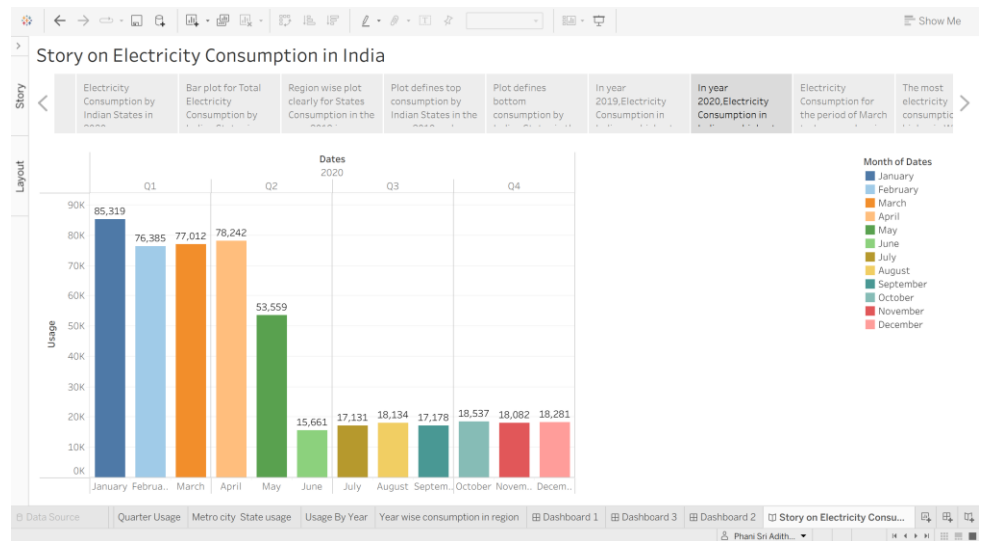
Story Point 9

Title

Electricity Consumption During Lockdown (March–June)

Description

This visualization compares electricity consumption during the lockdown months of March to June for both 2019 and 2020. It clearly demonstrates the impact of COVID-19 restrictions on electricity demand.



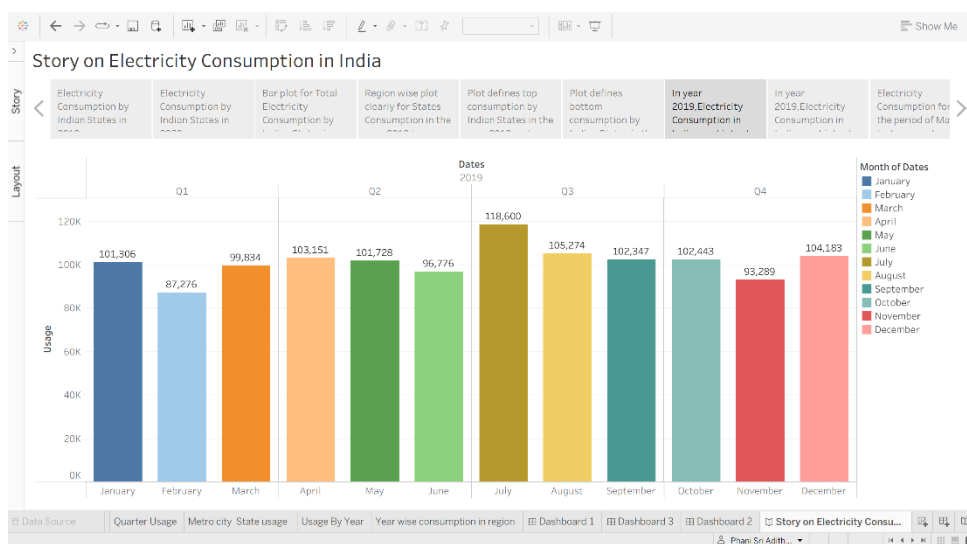
Story Point 10

Title

Monthly Electricity Consumption in 2020

Description

This chart illustrates monthly electricity consumption during 2020. A sharp decline is visible during the nationwide lockdown period between March and June.



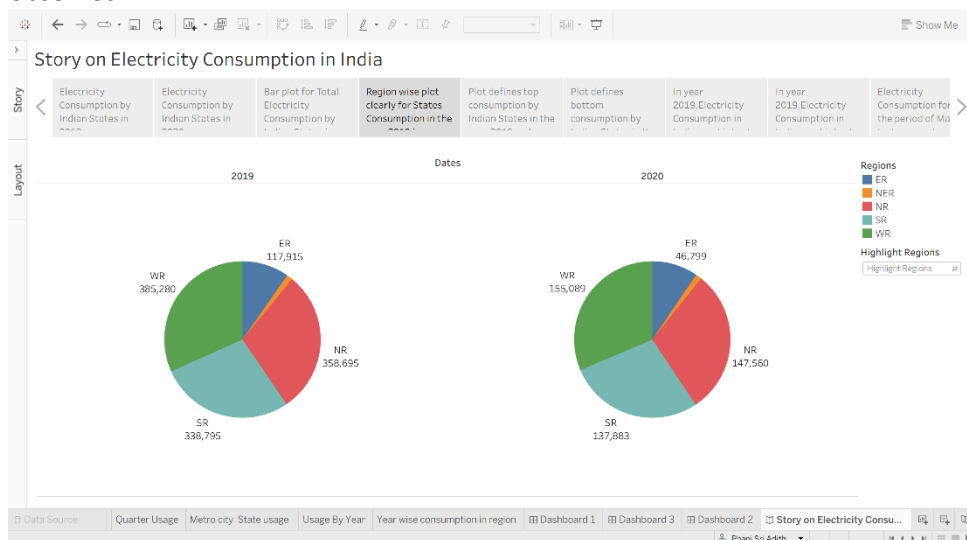
Story Point 11

Title

Monthly Electricity Consumption in 2019

Description

This visualization shows month-wise electricity consumption throughout 2019. Seasonal variations and increasing demand during mid-year can be clearly observed.



Story Point 12: Region-wise Electricity Consumption Comparison (2019 vs 2020)

Description:

This story presents a region-wise comparison of electricity consumption in India for the years 2019 and 2020 using pie charts. The visualization groups states into five geographical regions—Northern (NR), Southern (SR), Eastern (ER), Western (WR), and North-Eastern (NER)—to analyze regional electricity demand. The charts reveal that the Western Region (WR) recorded the highest electricity consumption in both years, while the North-Eastern Region (NER) had the lowest.

